STIR-L4 plans a systematic review with human input, through a staged, step-by-step process, with continuous awareness of the whole process as a system.
Three proposed representations, applied in turn, make the evidence behind planning traceable: Paper as an Object, Literature as a Tensor and Registering Evidence Dynamically.
Every value keeps its source and can be checked by a person or the model; a person's correction always overrides the model.

(a) Paper as an Object (PAO)

each paper is one object: three kinds of property and a ledger

Paper 3
Paper 2
Paper 1 one object

1 Identity *immutable*
DOI · PMID · PMCID · OpenAlex ID · title
authors · year · abstract · retraction status
used to merge duplicates and link reports of one study

2 Standard properties *verifiable*
publication type · affiliations · funding
study design · sample size · setting · country
the same fields in every review

3 Custom data extraction *for this review*
drug: empagliflozin · comparator: placebo
population: adults without diabetes
outcomes: fat mass, lean mass (DXA scan)
result: change at week 24, for each outcome

4 Ledger *the evidence for each value*

value	fat mass, week 24: −1.1 kg
source	*"…fell by 1.1 kg…" (Results, p. 5)*
status	reported (or not reported, unclear)
checked	the model extracted it; Uday verified it

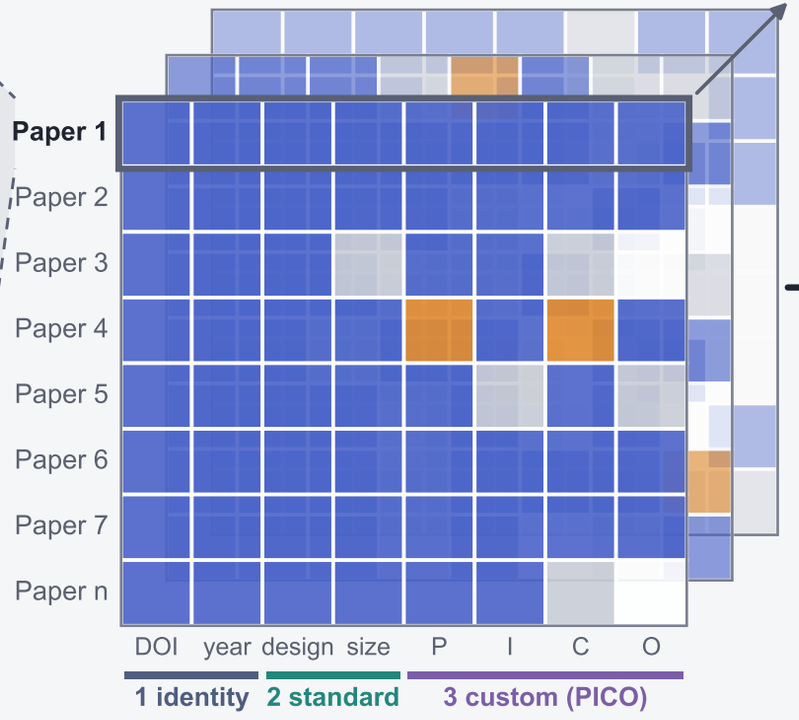
extract → verify → cite → ask a person
a person's correction overrides the model

(b) Literature as a Tensor (LAT)

each paper becomes one row; layers add a third property

one paper → one row

third property, e.g., region: one layer per level



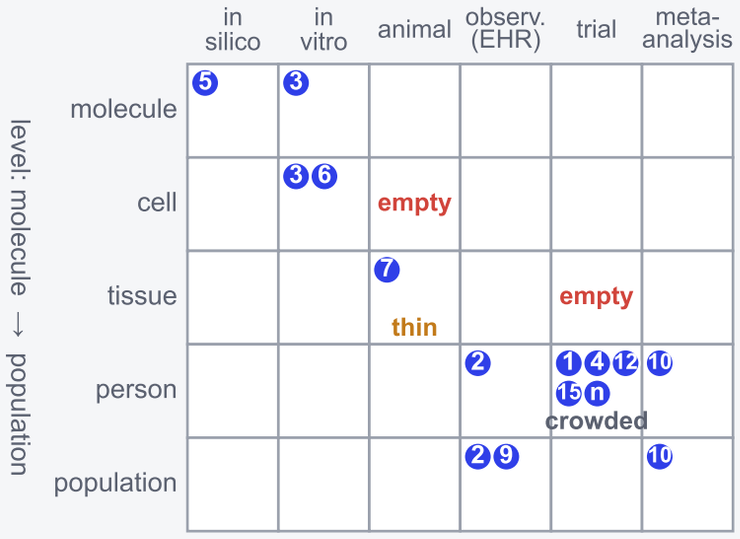
reported · not reported · conflicting · unclear

Each cell is one value with its ledger (panel a, block 4).
Duplicates are merged first, so each study is counted once.
Columns and their levels are fixed before the search.
Layers add a third property (for example region or year),
so candidate gaps that two-way tables hide become visible.

(c) Registering Evidence Dynamically (RED)

the landscape of the field: every kind of study on one atlas

The atlas: one topic, every kind of study
drawn before the search; each circle is a paper from (b)



Placing a paper, as in image registration
1 by rule, from its properties (e.g., its design)
2 by how closely its text matches a section
3 by a person, when neither settles it
a paper can sit in more than one section (2, 3, 10)

Reading the atlas
thresholds: empty 0 · thin 1–2 · crowded 5 or more
an empty section is a candidate gap; a targeted search
speaks only for the sources and dates searched;
what the atlas shows returns to feasibility